1 A student investigates springs made of metal wire, as shown in Fig. 1.1.

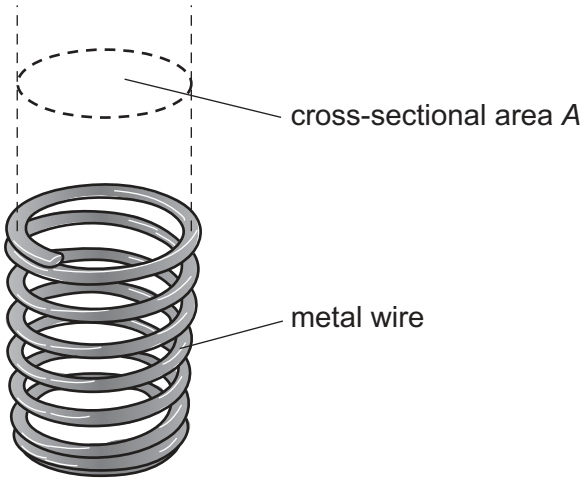

Fig. 1.1

The student constructs several springs from wire of thickness t. Each spring has a different cross-sectional area A.

The student investigates how the spring constant k varies with A.

It is suggested that the relationship between k and A is

$$k = \frac{\beta\rho t^4}{A^{\frac{3}{2}}N}$$

where ρ is the density of the metal, N is the number of turns of wire on the spring and β is a constant.

Design a laboratory experiment to test the relationship between k and A.
Explain how your results could be used to determine a value for β.

You should draw a diagram, on page 3, showing the arrangement of your equipment.
In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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